AUTOMATED BIOMASS ESTIMATION USING SELF-SUPERVISED VISION TRANSFORMERS

Research proposal: MEng

Candidate
Aaron Masuba

Student number
25737806

Department of Electrical, Electronic and Computer Engineering
University of Pretoria

13 May 2025

Supervisor
Prof Herman Myburgh
Dr Allan De Freitas

(Signature) Aaron Masuba
Candidate Aaron Masuba Date 13/05/2025

(Signature)
Supervisor/Promoter Date 14/05/2025

(Signature)
Postgraduate Committee Date 2 June 2025

I will approve this proposal, but I have serious reservations about the proposed scope (4 hypotheses) and output (3 journal papers). This would indicate a study with the scope and complexity of a PhD, and poses significant risk of delayed completion of the M.Eng. I would urge the candidate and supervisors to reconsider the scope of the study.

-Tinus Stander.

Summary Page

1. Research questions

- Can self-supervised vision transformer models be leveraged to learn rich visual representations of plants that capture both geometric structure and physiological traits relevant to biomass estimation?
- What transformer-based grounding and fusion strategies enable accurate 3D reconstruction of plant architecture from 2D RGB or RGB-D imagery?
- Can the reconstructed 3D plant representation be utilized to predict biomass in a label efficient manner, and what is the relationship between reconstruction model and biomass estimation accuracy?

2. Approach

This research proposes a framework that combines self-supervised vision transformer architectures with geometry-grounded 3D reconstruction for automated biomass estimation. The study will dive into multiple self-supervised Vision Transformer backbones to perform recognition and structural feature encoding. In addition to that, advanced transformer-based 3D reconstruction models will be evaluated for generating plant geometry from RGB-D and multiview images. Multimodal fusion techniques will integrate 2D semantic and 3D structural features for regression -based biomass prediction. In so doing, the study shall focus on data efficiency and generalizability with validation against ground-truth metrics such as MAE, RMSE, MARE, and Chamfer Distance.

3. Contribution

The study will contribute a self-supervised, label-efficient pipeline for plant biomass estimation that integrates state-of-the-art transformer models in both 2D representation learning and 3D geometry fusion. This bridges the gap between vision-based phenotyping and plant physiological modelling, advancing the scalability and generalizability of non-invasive biomass prediction tools in precision agriculture and environmental monitoring.

4. Anticipated peer-reviewed journal articles

Master's degree students

Target journal: International Journal of Computer Vision
Preliminary title: Representation Learning and Geometry Fusion for Generative 3D Reconstruction and Estimation
Description: This survey shall dive in literature review of both classical and modern approaches for generative 3D reconstruction and estimation. It shall look at studies using representation learning where learned features and priors facilitate reconstruction and geometry fusion where multi-view and or multi-modal information is combined.

Target journal: International Journal of Computer Vision
Preliminary title: Self-Supervised Vision Transformers for Semantic Reconstruction and Biomass Estimation from Multiview Plant Imagery
Research gap that will be addressed: Lack of label-efficient frameworks for integrating plant species recognition and 3D reconstruction using self-supervised transformers. The aim is to study how visual priors can drive biomass estimation in unstructured, real-word data without dense annotation.

Target journal: PFG-Journal of Photogrammetry, Remote Sensing and Geoinformation Science
Preliminary title: Multimodal Geometry Fusion for Automated Biomass Estimation in Precision Agriculture.
Research gap that will be addressed: Leveraging a developing field of geometry-aware fusion strategies for biomass estimation. Let say, what relationship can co-exist between 3D shape priors and physiological plant traits. The demonstration shall be based on how learned geometry from RGB or RGB-D imagery can inform biomass regression in various crops.

Automated Biomass Estimation using Self-Supervised Vision Transformers

Aaron Masuba

***Abstract*– Above-ground biomass (AGB) estimation is crucial for carbon accounting and climate change monitoring, as well as for agricultural productivity and plant health. Non-invasive methods leveraging LiDAR sensing and deep learning have emerged to predict AGB without destructive sampling. Recent models demonstrate progress using 3D point clouds and multi-modal imagery, respectively, but they remain limited by extensive data requirements, rigid model architectures, and poor transferability across heterogeneous vegetation and sensing conditions. To overcome these limitations, this research proposes a self-supervised learning framework for above-ground biomass estimation and plant species recognition. The approach hypothesises combining multi-view 2D-to-3D reconstruction of plant geometry from RGB-D imagery with state-of-the-art vision transformers. By exploiting geometry-aware, label-efficient training, the model will learn to recognize and reconstruct 3D plant structures from multi-angle RGB-D data. The framework will be evaluated on a diverse pool of datasets including RGB-D datasets captured with Intel RealSense cameras, assessing 3D reconstruction accuracy, species classification, and biomass estimation performance. By reducing reliance on manual labels and explicitly encoding 3D structural cues, the proposed approach aims to improve generalization across diverse plant types and environmental conditions. This work is expected to enable scalable, cost-effective biomass monitoring for precision agriculture, environmental modelling, and sustainable land-use planning.**

***Key words–Biomass estimation, deep learning, visual geometry, vision transformers, self-supervised learning*.**

I. Introduction

Automated estimation of plant biomass is essential for environmental monitoring, carbon accounting, and sustainable land-use management [1], [2], [3], [4], [5], [6], [7]. Some plants such as trees are natural carbon sinks and fundamental in regulating the Earth's climate [8]. Biomass estimation enables us to accurately quantify carbon storage, model tree growth and implement sustainable use case strategies for land[9], [10]. For crops, biomass enable us understand plant physiological conditions and monitor the growth cycles for effective planning [11]. Remote sensing technologies, particularly LiDAR (Light Detection and Ranging), have emerged as powerful tools for capturing detailed structured information from vegetation, enabling non-invasive, scalable biomass modelling [1], [2], [3], [7], [9], [12], [13], [14]. However, current LiDAR-based approaches often suffer from limitations in generalizability, and efficiency especially when deployed to heterogenous tree structures [15], [16], [17].

Accurate non-destructive estimation of plant biomass remains a challenge due to limitations in tradition and current deep learning approaches. Conventional biomass measurement requires destructive harvesting, which is labour-intensive, unscalable and precludes continuous monitoring of the same plant [15]. Imaging-based methods offer a faster alternative, but existing models often struggle to generalize across different species, growth condition, and experiments [15], [16], [18]. Traditional image-derived models often rely on simple proxies like projecting 2D area or vegetation indices and assume traits which can lead to large errors under varying physiological conditions [19]. Deep learning with Convolutional Neural Networks (CNNs) has improved image-based biomass prediction, but CNN-based pipelines also exhibit important shortcomings [20]. They typically require extensive labelled data and sometimes dense annotations such as pixel-wise segmentation and multi-view 3D scans [21], [22]. That limits scalability to new plant types or environments. To mention, standard CNNs process images in a localized manner and may not fully exploit global attention cues, making it hard to infer volumetric structure from a single view [4], [19], [8], [23]. 3D features that are important such as plant height, volume, and branching structure are therefore not explicitly learned, which limits the model's ability to distinguish plants with similar 2D appearance but different architecture [8]. Additionally, important physiological differences may not be captured by colour and or texture alone [24]. Existing CNN approaches usually first segment the plant from the background and then compute features for biomass regression [11]. This therefore propagates errors and struggle with occlusion from overlapping neighbouring plants affecting reliability of biomass estimations [18].

A recent CCN study [11] investigates a hybrid crop mapping approach integrating machine learning such as Support Vector Machines (SVM), Random Forest (RF) and Deep Learning (DL) techniques with Sentinel-1 and Sentinel-2 satellite imagery in Ardabil, Iran, a region known for its diverse crop types. Their framework employed object-based image analysis, multiscale segmentation, and Random Forest-based feature selection to optimize the classification process. The study found that while the CNN algorithm outperformed individual SVM and RF classifiers, the most accurate results were achieved by combining the outputs of all three algorithms using a majority voting method. Specifically, the combined approach yielded an overall accuracy of 90.34% and a Kappa coefficient of 88.72%, surpassing the CNN's individual performance (88.03% and 85.99%, respectively). The research also highlighted the importance of specific Sentinel-2 bands (red-edge, near-infrared, short-wave infrared) and the synergistic use of optical and radar data for improved crop separation. This study demonstrates the potential of hybrid machine learning and deep learning frameworks, coupled with data fusion techniques, to enhance the accuracy of crop mapping initiatives. However, this still suffers from lacking generalizability where it can be applied on a new species without retraining and calibrating for each new species or conditions.

In efforts to address some of these challenges, Vision Transformer-based solutions have quickly emerged [25], [26]. Vision Transformers (ViTs) provide a flexible architecture to encode global image context and integrate multiple information sources, which is helpful in modelling complex plant traits showing great promise in 2D vision tasks and are rapidly being adapted for 3D scene understanding [27], [28], [29]. However, their application in plant ecological environments including trees and crops, especially for biomass estimation using point clouds or RGB-D images, remains underexplored [8]. The use of 3D point cloud data for biomass estimation has traditionally relied on extracting empirical features such as tree height, crown diameter and canopy volume [4], [6], [8], [30], [31], [32]. Classical approaches incorporate regression models or allometric equations to infer biomass[33]. While effective in controlled settings, these techniques falter in generalizing to diverse species and varying acquisition conditions.

In the study [34] proposed an end-to-end deep learning network for above-ground biomass (AGB) estimation from LIDAR point clouds, comprising a pillar-based point feature encoder and an MLP-Mixer module for aggregated feature learning. Drawing inspiration from 3D object detection techniques and MLP-mixer architecture from 2D computer vision, the method first groups the point cloud into vertical pillars and extracts point-level features via a simplified PointNet-like encoder. It then applies MLP-Mixer blocks to mux token and channel information across pillars, culminating in an AGB prediction via a final fully connected layer. When applied to small-grain cereal (wheat) fields plots for non-destructive, high-throughput biomass phenotyping, BioPM achieved competitive performance on the SGCBP dataset, significantly outperforming a classical LiDAR-based biomass approach (3DVI) by approximately 33% in mean absolute error and reaching parity with more complex deep learning models such as BioNet [7]. Notably, BioPM also outperformed BioNet variant without point-cloud completion by approximately 19%, demonstrating robust performance even without explicit point cloud completion. Moreover, its streamline architecture enables fast inference. However, the approach required data augmentation to compensate for limited training data, and ad-hoc handling of variations in canopy structure (e.g., duplicating sparse flowing-stage point cloud data) was needed, indicating challenges in generalizing across different growth stages. While not explicitly evaluated for broader 3D reconstruction or other species, the reliance on pillar grouping may incur information loss and would likely require further model enhancements to handle different crop structures or phenotyping tasks beyond biomass estimation. Nonetheless, BioPM demonstrates a promising and efficient approach for LiDAR-based biomass prediction in agriculture.

The study [18] introduces NeFF-BioNet a novel deep learning framework for crop biomass prediction adaptable to both point cloud and drone imagery data. The core BioNet leverages a sparse 3D CNN and a transformer encoder to extract and process 3D features, achieving state-of-the-art performance in biomass prediction from point clouds, with a 6.1% relative improvement (RI) over existing methods. To extend the framework to RGB imagery, the authors integrate a

Neural Feature Field (NeFF) module, enabling 3D structure reconstruction and semantic feature extraction from drone images. The NeFF-BioNet combination demonstrates a 7.9% RI in biomass prediction from drone imagery, offering a scalable solution for large-scale agricultural applications. Performance is evaluated using Mean Absolute Error (MAE), Mean Absolute Relative Error (MARE), and Root Mean Square Error (RMSE), with the model consistently outperforming benchmarks and demonstrating the efficacy of its hybrid 3D CNN and transformer architecture. The study highlights the potential of vision-based biomass estimation using affordable drone-mounted cameras.

Novel techniques involving self-distillation with no labels have been developed to help with scalability using a teacher student model that learns from un-labelled or lightly labelled data [35], [36]. WaveFormer introduced a 3D Transformer that incorporates Discrete Wavelet Transform (DWT) to split features into frequency bands, allowing efficient context modelling and detail preservation [27]. This biologically inspired approach mimics human visual pathways and has proven effective in volumetric segmentation of medical scans. Visual Geometry Grounded Transformer (VGGT) applied grounding attention layers in spatial geometry [28]. VGGT learns 3D scene reconstructions from single or multi-view and demonstrates that geometric inductive biases can improve accuracy and generalizability in vision tasks. Despite these innovations, few studies advanced on these developments to remodel biomass estimation not filling the gap of robustness, label efficiency and approaches of rich visual cues to model plant biomass in a general way.

Leveraging the existing body of science, this proposal suggests following a first principles design approach of biomass estimation using computer vision embedding the 3R's that include recognition, reorganization and reconstruction and apply to RGB or RGB-D imagery of plants [37]. Benchmarking with various transformer backbones and self-supervision from 2D images, encoding essential plant structures without requiring dense labels shall be explored [26], [27], [35], [36], [38], [39], [40], [41], [42]. The self-supervised objectives like predicting masked image patches and contrastive view alignment shall be exploited to challenge the model to absorb universal visual patterns in plants, yielding representations that are grounded in both frequency and spatial domains. Frequency grounding means the model is encouraged to capture information across different scales and textural frequencies for example broad shapes of fine-grained leaf textures while geometry grounding means incorporating spatial cues like the multi-view perspectives and depth information to learn features corresponding to real 3D structure. Following that, learned features will be reorganized and fused within the transformer to form a coherent latent representation of the plant. The ViT's self-attention mechanism enables fusion of multimodal inputs including combining colour and depth channels to multiple camera views ensuing that information about the plant's geometry is integrated with its appearance. This reorganization step shall be important in a way that it will assemble a holistic view of the plant aligning features that correspond to the same physical points and regions in space. Lastly, the proposal shall proceed with a pipeline enriched representation to reconstruct a 3D model of the plant and estimate its biomass. Unlike the dense reconstruction, this suggests predicting structural properties like 3D point clouds, voxel grid and parametric volume from the transformers features from which biomass can be inferred by volume regression [43]. If the biomass is predicted from the reconstructed 3D structure instead of directly from 2D pixel intensities, the integrated 3R approach shall be designed to be modular automating each stage, label efficient by learning from self-supervised mechanism requiring a small set of biomass labelled samples for final calibration and generalizable aiming at transformers learned representations to transfer across species and environments.

II. BIOMASS ESTIMATION USING SELF-SUPERVISED VISION TRANSFORMERS.

A. *Research Questions*

The research questions are as follows.

1) Can self-supervised vision transformer models be leveraged to learn rich visual representations of plants that capture both geometric structure and physiological traits relevant to biomass estimation?

   This question aims to identify what self-supervised strategies a ViT can utilise to learn plan features that generalise across different types with minimal labelled data.

2) What transformer-based grounding and fusion strategies enable accurate 3D reconstruction of plant architecture from 2D RGB or RGB-D imagery?

   This question explores ways to reorganise information for 3D understanding for example how spatial geometry cues and frequency multi-scale information can be integrated into the transformer's attention mechanism to fuse multiple views or modalities to yield coherent 3D plant model.

3) Can the reconstructed 3D plant representation be utilized to predict biomass in a label-efficient manner, and what is the relationship between reconstruction model and biomass estimation accuracy?

   The interest here lies on the final estimation given a transformer-derived 3D structure, to say what learned mappings best translate that structure into an accurate biomass prediction and whether reconstructing geometry leads to improved biomass estimates especially under limited training labels compared to direct 2D image-to=biomass prediction.

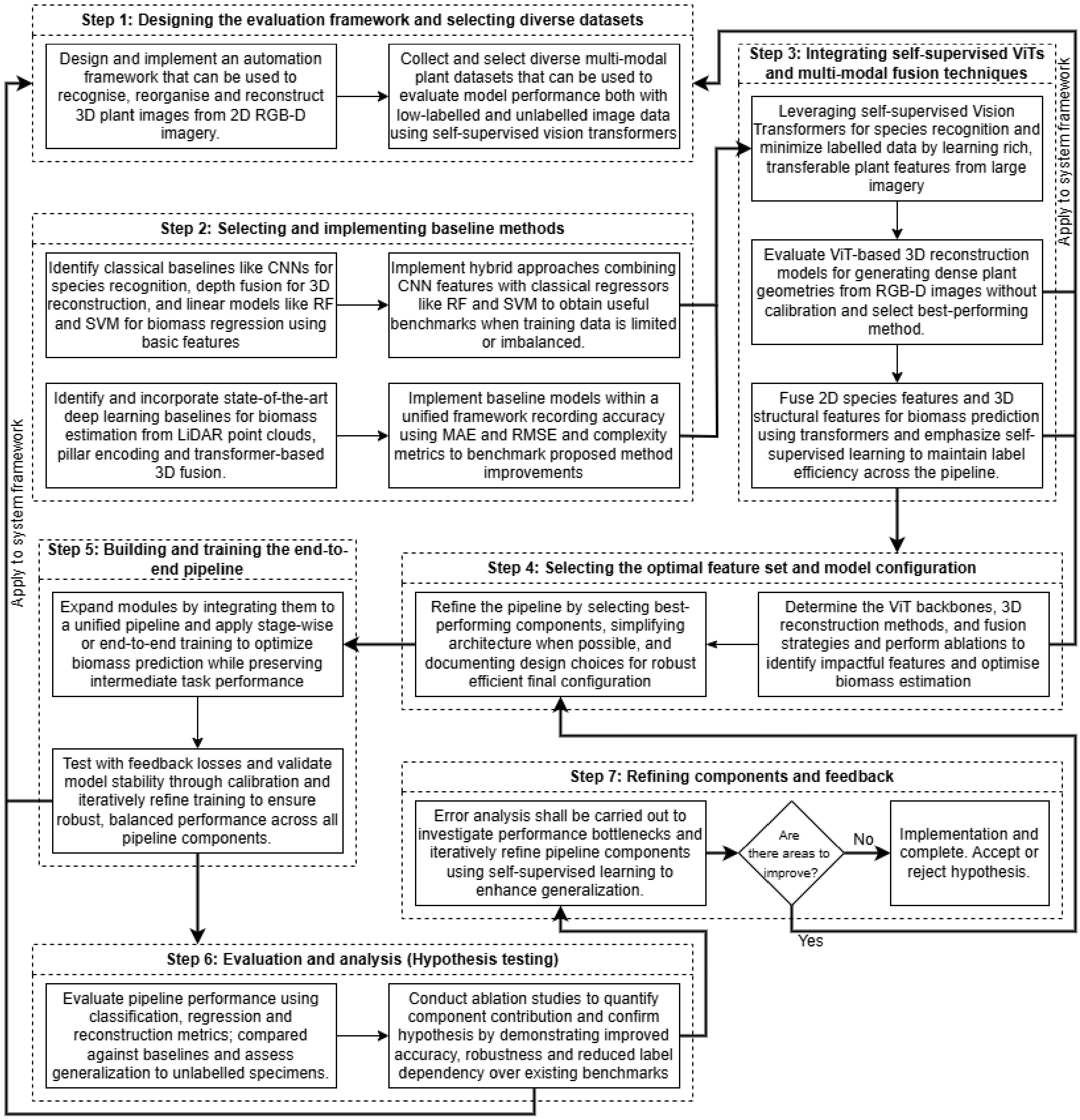
B. Hypotheses

1. Vision Transformers enhanced with recognition and reorganisation will outperform traditional CNNs in segmentation accuracy and biomass prediction. The use of self-supervised vision transformers can learn generalizable plant feature representations that encode 3D structural and physiological information, enabling accurate biomass prediction with substantially fewer labelled examples than fully supervised approaches.
2. Geometry-grounded models will show higher consistency across viewpoints and better 3D reconstruction quality. When a transformer is enriched with geometry grounding, it may effectively fuse multi-view or multi-modal inputs to produce faithful 3D reconstruction of a plant capturing traits like volume, branch architecture and leaf density.
3. The combination of frequency and geometry grounding improves parameter efficiency especially as relates to positional encodings for 3D spectral features. This follows that even in presence of inductive biases like positional encoding for 3D coordinates or the spectral features into a Vision Transformer, the reconstructions shall closely reflect the plant's true physical structure even where some parts might be occluded.
4. Hybrid models will demonstrate superior robustness to occlusion, noise, and sparse sampling conditions in RGB camera scans. The expectation is for vision transformers to provide reliable basis for estimating biomass and generalizable for different imaging conditions.

C. Proposed Research Methodology

To answer the research questions outlined above, a research structure intended to be used has been created. Figure 1 below shows an overview of the intended research procedure.

1) *Designing the evaluation framework and obtaining data*: To ensure consistent benchmarking of plant species recognition (classification), 3D plant reconstruction from RGB-D and or multi-view data from our camera rig system and biomass estimation (regression), an evaluation framework is proposed. The framework suggests performing the three tasks making use of standard metrics such as classification accuracy for species identity, 3D reconstruction fidelity like Intersection of Union (IoU) with ground truth-shapes and biomass prediction error from Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Relative Error (MARE). Once the framework has been implemented, a set of appropriate datasets encompassing species variations, growth stages and structure variations shall be sought. These shall be split into training, validation and hold-out test sets, with some species of environments reserved exclusively for testing to evaluate generalization. The most important note shall be ensuring that dataset includes necessary modalities such as RGB images for species identity and 2D features. This is important to ensure the body of science is contributed to and the solution can be reproduced for verification.



2) *Selecting and implementing baseline methods*: Before the development of proposed system commences, identification of classical pipelines as used in prior work for biomass traits will lay basis for activities such as recognition, reconstruction and estimation of biomass. These shall serve as point of comparison and baseline for proposed system to assess the viability species classification task, running 3D reconstructions and biomass estimation with baseline models using the available training and test set. These baseline results and their complexity will contextualize the improvements of the proposed method and are required to ensure if the baseline systems indicate better performance proposed self-supervised vision transformer framework.

3) *Integrating self-supervised ViTs and multimodal fusion techniques*: Once the baseline systems have been put in place and implemented, it will be expected that vision transformer (ViT) can learn generic plant structures and textures through self-supervision requiring minimal labelling and 3D reconstruction using RGB or RGB-D inputs in a self-supervised or weakly-supervised

manner will be possible. This shall be derived from various existing models with state-of-the-art performance but not generalized to plant morphology and applications such as biomass estimation. By determining the best performing self-supervision learning models and 3D reconstruction ViT-based model, a feature fusion technique shall follow applying the realized concatenated methods to the baseline implementations. The selected approach shall be iterated and tested in evaluation framework using the datasets in step 1 to verify that each part produces plausible outputs before full optimization.

4) *Selecting the optimal feature set and model configuration*: Having implemented a baseline of fused-feature extractor with 3D reconstruction and fusion regressor, systematic experiments are proposed to identify the most effective architecture. These experiments shall be conducted on different transformer backbones and fusion strategies linking 2D features with 3D structural representations. Each configuration will be assessed using the defined evaluation framework to determine its contribution to biomass prediction accuracy. Feature ablations will quantify the importance of each modality and guide refinement. There on, the performance from the most optimal components will be selected to guide documentation and incorporation into the final system configuration. In doing so, this will be important to ensure that the selected features and design choices are empirically justified, valid across tasks and supported by rigorous experimentation.

5) *Building and training the end-to-end pipeline*: Following the selection of optimal components, this phase proposes to involve integrating the species recognition, 3D reconstruction, and biomass estimation modules into a unified end-to-end framework. Careful consideration shall be given to the training strategy and adopting either a stage-wise or end-to end fine-tuning to preserved learned representations and maximize performance. A loss function for multi-task might be employed to maintain balanced learning across tasks. Auxiliary constraints such as reconstruction consistence or biomass-volume alignment might guide the refinement of earlier modules while training. The study proposes a validation of the pipeline to ensure each component maintains expected performance without interference. Iterative calibration and detailed logging of convergence behaviour, overfitting and other dynamics will support informed adjustment and ensure training stability. The intention shall be to extract a final output that is fully trained and validated for biomass estimation which is structurally aligned.

6) *Evaluation and analysis to litmus-test the hypothesis*: The final pipeline will be rigorously evaluated against held-out test sets, using defined metrics such as MAE, RMSE, MARE, and reconstruction fidelity scores. The performance will be benchmarked against classical, deep learning and hybrid baselines. Generalizability litmus test on new species with label efficiency will be assessed through subsampled training experiments. To follow, ablation studies will isolate the impact of each module from species recognition, 3D reconstruction, and self-supervised

features on overall accuracy. The combination shall be necessary in validating the hypothesis that a self-supervised ViT-based pipeline improves performance, robustness and scalability over the state-of-the-art methods.

7) *Iterative refining and feedback*: Based on obtained results, final refinements will be guided by detailed error analysis, targeting misclassifications, reconstruction artifacts, and systemic biases. The identified issues will prompt targeted improvements in species recognition, 3D reconstruction and biomass regression with potential use of data augmentation, model tuning and expanding pretraining. The research proposes that a modular design will enable updates to specific components without full re-training. The insights from the process will be cycled back into earlier pipeline stages to test alternate fusion strategies and learning objectives at which final point, the research shall be revised to accept or reject the hypothesis.

III. CONCLUSION

This research proposal suggests a methodological breakthrough in automated above-ground biomass estimation by introducing a self-supervised vision transformer framework that leverages the 3Rs of computer vision [37]. The research suggests integrating these three pillars, the approach shall learn rich, holistic representations from RGB(-D) imagery, enabling accurate and generalizable biomass predictions with minimal labelled data. One of the anticipated outcome and scientific contribution of this work shall result into a novel biomass estimation framework using new deep learning frameworks that combine recognition, reorganization and 3D reconstruction tasks to estimate above ground biomass. This self-supervised vision transformer model is expected to achieve state-of-the-art accuracy while maintaining robust performance across diverse plant traits improving model generalizability. The second anticipated outcomes and scientific contribution shall be improving label efficiency through demonstrating self-supervision multi-task learning to drastically reduce the need for extensive labelled training data. The exploitation of unlabelled images for pretraining and auxiliary tasks will enable learning transferable features from biomass estimation enhancing label efficiency without sacrificing accuracy. The third outcomes shall be benchmarking exploration of 3D reconstruction from 2D data by introducing techniques that infer 3D structural data from 2D RGB imagery augmented with depth depending on camera modules. This shall contribute to a gap between 2D computer vision and volumetric understanding of plants improving reconstruction fidelity and providing insights that might benefit broader 3D reconstruction and computer vision research. At the end of the project, the research looks forward to open source the datasets and tools to strengthen the commitment towards open science by releasing the curated datasets and complete implementation of the proposed framework to enable reproducibility of results and encourage follow-up work given alignment with university guidelines on research distribution.

Looking beyond the direct outcomes and body of science contribution, the proposed work is of significant impact in agriculture through deployment of rapid non-invasive biomass estimation frameworks that will potentially transform precision agriculture. Above-ground biomass is critical for understanding crop performance that directly affects yield and informs farm management decision. The tools developed in the process of carrying out this research shall enable farmers and agronomists to quickly assess crop growth and productivity in the field. In lieu of the environment, biomass estimation offers insights on how to manage and sustain ecosystem health and vitality sequestrating carbon and modelling climate change [15]. Lastly, reconstruction of complex 3D structures from visual data shall be a significant milestone especially when efficiency and multi-view consistency are embodied into the natural environments. This shall unlock broader opportunities of 3D reconstruction application such as improving digital twin models of crops and enhancing robotic perception in nature.

In the eve of pursuing this proposed topic and methodology for the above outcomes, it is necessary to point out several key challenges implicit to this field. Model generalizability remains a foremost concern where deep models often struggle when applied to unseen plant species or novel conditions and this means rigorous validation and techniques like domain adaptation and diverse training data will be needed to ensure robust performance[44]. Label efficiency is another challenge from obtaining ground-truth biomass at scale which is labour-intensive making the self-supervised approach critical to minimize labelled data requirements through careful tuning to maintain accuracy with limited labels [35], [40]. Data diversity is necessary to reduce bias and thus a system able to handle various viewpoints and conditioning shall play a crucial role [36], [41]. Finally, achieving hight reconstruction fidelity from 2D imagery is an open challenge where factors such as occlusion and complex morphology of the plant or specimen being studied affect accuracy[28], [38], [39]. Therefore, the proposal claims to incorporate RGB-D data and multi-view imaging which might improve fidelity but overcoming the limits of inferring 3D biomass from limited images with require careful algorithmic design and thus shall remain a continued area for innovation. Clear understanding of these challenges shall guide the research methodology and experimentation ensuring that a solution developed is realistic and robust.

In summary, the proposal aims to advance the frontiers of computer vision and machine learning through novel application-driven framework, while simultaneously providing practical tolls for agriculture and environmental monitoring. The expected scientific contributions from improving vision transformer generalization to enabling label-efficient biomass estimation will lay a strong foundation for future research at the intersection of artificial intelligence and agricultural engineering. Working on the challenges above, this work shall improve accuracy and generality of above-ground biomass estimation, which has broad implications for sustainable farming, ecological assessment and 3D vision. The findings of automated biomass estimation using self-supervised vision transformers shall be circulated in research journals, articles and conference proceedings to ensure that the benefits are widely realized across both the academia and industry.

References

[1] A. Ruhan, W. Du, H. Ying, B. Wei, Y. Shan, and H. Dai, "Estimation of Aboveground Biomass of Individual Trees by Backpack LiDAR Based on Parameter-Optimized Quantitative Structural Models (AdQSM)," *Forests*, vol. 14, no. 3, 2023, doi: 10.3390/f14030475.

[2] B. Aricak, M. G. Wing, and A. E. Akay, "State of the Art on Airborne LiDAR Applications in the Field of Forest Engineering," in *Concepts and Applications of Remote Sensing in Forestry*, 2023. doi: 10.1007/978-981-19-4200-6_18.

[3] L. Zhou *et al.*, "Estimating 3D Green Volume and Aboveground Biomass of Urban Forest Trees by UAV-Lidar," *Remote Sens (Basel)*, vol. 14, no. 20, 2022, doi: 10.3390/rs14205211.

[4] Q. Song, Y. Wang, and X. X. Zhu, "3d Point Cloud Simulation for Above-Ground Forest Biomass Estimation," in *International Geoscience and Remote Sensing Symposium (IGARSS)*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 1357–1360. doi: 10.1109/IGARSS52108.2023.10282570.

[5] S. Oehmcke *et al.*, "Deep Learning Based 3D Point Cloud Regression for Estimating Forest Biomass," Dec. 2021, [Online]. Available: http://arxiv.org/abs/2112.11335

[6] Q. Song, Y. Wang, and X. X. Zhu, "3d Point Cloud Simulation for Above-Ground Forest Biomass Estimation," in *International Geoscience and Remote Sensing Symposium (IGARSS)*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 1357–1360. doi: 10.1109/IGARSS52108.2023.10282570.

[7] L. Pan *et al.*, "Biomass Prediction with 3D Point Clouds from LiDAR."

[8] S. Oehmcke *et al.*, "Deep Learning Based 3D Point Cloud Regression for Estimating Forest Biomass," Dec. 2021, [Online]. Available: http://arxiv.org/abs/2112.11335

[9] N. Fedorov *et al.*, "Estimation of Carbon Stocks of Birch Forests on Abandoned Arable Lands in the Cis-Ural Using Unmanned Aerial Vehicle-Mounted LiDAR Camera," *Forests*, vol. 14, no. 12, 2023, doi: 10.3390/f14122392.

[10] N. Fedorov *et al.*, "Estimation of Carbon Stocks of Birch Forests on Abandoned Arable Lands in the Cis-Ural Using Unmanned Aerial Vehicle-Mounted LiDAR Camera," *Forests*, vol. 14, no. 12, 2023, doi: 10.3390/f14122392.

[11] B. Asadi and A. Shamsoddini, "Crop mapping through a hybrid machine learning and deep learning method," *Remote Sens Appl*, vol. 33, Jan. 2024, doi: 10.1016/j.rsase.2023.101090.

[12] J. Gonzalez de Tanago *et al.*, "Estimation of above-ground biomass of large tropical trees with terrestrial LiDAR," *Methods Ecol Evol*, vol. 9, no. 2, 2018, doi: 10.1111/2041-210X.12904.

[13] B. Aricak, M. G. Wing, and A. E. Akay, "State of the Art on Airborne LiDAR Applications in the Field of Forest Engineering," in *Concepts and Applications of Remote Sensing in Forestry*, 2023. doi: 10.1007/978-981-19-4200-6_18.

[14] L. Zhou *et al.*, "Estimating 3D Green Volume and Aboveground Biomass of Urban Forest Trees by UAV-Lidar," *Remote Sens (Basel)*, vol. 14, no. 20, 2022, doi: 10.3390/rs14205211.

[15] T. Ma *et al.*, "Development of forest aboveground biomass estimation, its problems and future solutions: A review," Feb. 01, 2024, *Elsevier B.V.* doi: 10.1016/j.ecolind.2024.111653.

[16] F. Wang *et al.*, "Development of Estimation Models for Individual Tree Aboveground Biomass Based on TLS-Derived Parameters," *Forests*, vol. 14, no. 2, 2023, doi: 10.3390/f14020351.
[17] F. Wang and M. Bryson, "Tree Segmentation and Parameter Measurement from Point Clouds Using Deep and Handcrafted Features," *Remote Sens (Basel)*, vol. 15, no. 4, Feb. 2023, doi: 10.3390/rs15041086.
[18] X. Li *et al.*, "NeFF-BioNet: Crop Biomass Prediction from Point Cloud to Drone Imagery," Oct. 2024, [Online]. Available: http://arxiv.org/abs/2410.23901
[19] A. Weiskittel *et al.*, "GAPS IN SAMPLING AND LIMITATIONS TO TREE BIOMASS ESTIMATION: A REVIEW OF PAST SAMPLING EFFORTS OVER THE PAST 50 YEARS," DW.
[20] D. Han, Q. Liu, and W. Fan, "A new image classification method using CNN transfer learning and web data augmentation," *Expert Syst Appl*, vol. 95, pp. 43–56, Apr. 2018, doi: 10.1016/j.eswa.2017.11.028.
[21] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," Sep. 2014, [Online]. Available: http://arxiv.org/abs/1409.1556
[22] S. Zagoruyko and N. Komodakis, "Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer," Dec. 2016, [Online]. Available: http://arxiv.org/abs/1612.03928
[23] Z. Xi, C. Hopkinson, and L. Chasmer, "Supervised terrestrial to airborne laser scanner model calibration for 3D individual-tree attribute mapping using deep neural networks," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 209, 2024, doi: 10.1016/j.isprsjprs.2024.02.010.
[24] C. No and M. Lee, "Global and local feature communications with transformers for 3D human pose estimation," *Sci Rep*, vol. 15, no. 1, Dec. 2025, doi: 10.1038/s41598-025-91426-w.
[25] A. Vaswani *et al.*, "Attention Is All You Need," Jun. 2017, [Online]. Available: http://arxiv.org/abs/1706.03762
[26] A. Dosovitskiy *et al.*, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," Oct. 2020, [Online]. Available: http://arxiv.org/abs/2010.11929
[27] M. M. Al Hasan *et al.*, "WaveFormer: A 3D Transformer with Wavelet-Driven Feature Representation for Efficient Medical Image Segmentation," Mar. 2025, [Online]. Available: http://arxiv.org/abs/2503.23764
[28] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, "VGGT: Visual Geometry Grounded Transformer," Mar. 2025, [Online]. Available: http://arxiv.org/abs/2503.11651
[29] C. Yuan *et al.*, "A Survey of Graph Transformers: Architectures, Theories and Applications."
[30] L. Pan *et al.*, "Biomass Prediction with 3D Point Clouds from LiDAR."
[31] W. Wang *et al.*, "Branching the limits: Robust 3D tree reconstruction from incomplete laser point clouds," *International Journal of Applied Earth Observation and Geoinformation*, vol. 125, 2023, doi: 10.1016/j.jag.2023.103557.
[32] J. Wijesingha, T. Moeckel, F. Hensgen, and M. Wachendorf, "Evaluation of 3D point cloud-based models for the prediction of grassland biomass," *International Journal of Applied Earth Observation and Geoinformation*, vol. 78, pp. 352–359, Jun. 2019, doi: 10.1016/j.jag.2018.10.006.
[33] S. R. Rodda, R. R. Nidamanuri, T. Mayamanikandan, G. Rajashekar, C. S. Jha, and V. K. Dadhwal, "Non-Destructive Allometric Modeling for Tree Volume Estimation in Tropical Dry Deciduous Forests of India Using Terrestrial Laser Scanner," *Journal of the Indian Society of Remote Sensing*, vol. 52, no. 4, 2024, doi: 10.1007/s12524-022-01664-y.
[34] Y. Lei and H. Ma, "BioPM: Mixer for Point Cloud Based Biomass Prediction."

[35] R. G. Lopes, D. Ha, D. Eck, and J. Shlens, “A Learned Representation for Scalable Vector Graphics,” Apr. 2019, [Online]. Available: http://arxiv.org/abs/1904.02632
[36] M. Oquab *et al.*, “DINOv2: Learning Robust Visual Features without Supervision,” Apr. 2023, [Online]. Available: http://arxiv.org/abs/2304.07193
[37] J. Malik *et al.*, “The three R’s of computer vision: Recognition, reconstruction and reorganization,” *Pattern Recognit Lett*, vol. 72, pp. 4–14, Mar. 2016, doi: 10.1016/j.patrec.2016.01.019.
[38] V. Leroy, Y. Cabon, and J. Revaud, “Grounding Image Matching in 3D with MASt3R,” Jun. 2024, [Online]. Available: http://arxiv.org/abs/2406.09756
[39] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, “DUSt3R: Geometric 3D Vision Made Easy,” Dec. 2023, [Online]. Available: http://arxiv.org/abs/2312.14132
[40] M. Caron *et al.*, “Emerging Properties in Self-Supervised Vision Transformers,” Apr. 2021, [Online]. Available: http://arxiv.org/abs/2104.14294
[41] T. Darcet, M. Oquab, J. Mairal, and P. Bojanowski, “Vision Transformers Need Registers,” Sep. 2023, [Online]. Available: http://arxiv.org/abs/2309.16588
[42] J. Zhang *et al.*, “MONST3R: A SIMPLE APPROACH FOR ESTIMATING GEOMETRY IN THE PRESENCE OF MOTION.” [Online]. Available: https://monst3r-project.github.io/.
[43] Y. Huang, L. Liu, Y. Gao, H. Zhang, and H. Li, “D2S: Towards Efficient Sparse 3D Object Detection via Dense to Sparse Knowledge Distillation,” in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, Apr. 2025, pp. 1–5. doi: 10.1109/ICASSP49660.2025.10888352.
[44] X. Lv *et al.*, “Test-Time Domain Generalization via Universe Learning: A Multi-Graph Matching Approach for Medical Image Segmentation,” Mar. 2025, [Online]. Available: http://arxiv.org/abs/2503.13012

Contact Information

Postal address: Room 4-14 3, Jakaranda, University of Pretoria, 478 Festival St, Hatfield, Gauteng, 0083.
E-mail: aaron.masuba@tuks.co.za or aaron.masuba@gmail.com
Tel number: +27645542093 or +256774434600
Fax number: N/A
Cell number: +27645542093